ابتدا فایل دیتاست موردنظر را می خوانیم :

```
import pyspark
from pyspark.sql import SparkSession

spark = SparkSession.builder.getOrCreate()

# Read json file
news_df=spark.read.json("/content/gdrive/MyDrive/news/News_Category_Dataset_v3.json")
news_df.toPandas()
```

index	authors	category	date	headline	link	short_description
0	Carla K. Johnson, AP	U.S. NEWS	2022-09-23	Over 4 Million Americans Roll Up Sleeves For Omicron-Targeted COVID Boosters	https://www.huffpost.com/entry/covid-boosters-uptake-us_n_632d719ee4b087fae6feaac9	Health experts said it is too early to predict whether demand would match up with the 171 million doses of the new boosters the U.S. ordered for the fall.
1	Mary Papenfuss	U.S. NEWS	2022-09-23	American Airlines Flyer Charged, Banned For Life After Punching Flight Attendant On Video	https://www.huffpost.com/entry/american-airlines-passenger-banned-flight-attendant-punch-justice-department_n_632e25d3e4b0e247890329fe	He was subdued by passengers and crew when he fled to the back of the aircraft after the confrontation, according to the U.S. attorney's office in Los Angeles.
2	Elyse Wanshel	COMEDY	2022-09-23	23 Of The Funniest Tweets About Cats And Dogs This Week (Sept. 17-23)	https://www.huffpost.com/entry/funniest-tweets-cats-dogs-september-17-23_n_632de332e4b0695c1d81dc02	"Until you have a dog you don't understand what could be eaten."
3	Caroline Bologna	PARENTING	2022-09-23	The Funniest Tweets From Parents This Week (Sept. 17-23)	https://www.huffpost.com/entry/funniest-parenting-tweets_l_632d7d15e4b0d12b5403e479	"Accidentally put grown-up toothpaste on my toddler's toothbrush and he screamed like I was cleaning his teeth with a Carolina Reaper dipped in Tabasco sauce."
4	Nina Golgowski	U.S. NEWS	2022-09-22	Woman Who Called Cops On Black Bird-Watcher Loses Lawsuit Against Ex-Employer	https://www.huffpost.com/entry/amy-cooper-loses-discrimination-lawsuit-franklin-templeton_n_632c6463e4b09d8701bd227e	Amy Cooper accused investment firm Franklin Templeton of unfairly firing her and branding her a racist after video of the Central Park encounter went viral.
5		U.S. NEWS	2022-09-22	Cleaner Was Dead In Belk Bathroom For 4 Days Before Body Found: Police	https://www.huffpost.com/entry/belk-worker-found-dead-columbiana-centre-bathroom_n_632c5f8ce4b0572027b0251d	The 63-year-old woman was seen working at the South Carolina store on Thursday. She was found dead Monday after her family reported her missing, authorities said.
6	Elyse Wanshel	U.S. NEWS	2022-09-22	Reporter Gets Adorable Surprise From Her Boyfriend While Live On TV	https://www.huffpost.com/entry/reporter-gets-adorable-surprise-from-her-boyfriend-while-working-live-on-tv_n_632ccf43e4b0572027b10d74	"Who's that behind you?" an anchor for New York's PIX11 asked journalist Michelle Ross as she finished up an interview.
7	DÁNICA COTO, AP	WORLD NEWS	2022-09-22	Puerto Ricans Desperate For Water After Hurricane Fiona's Rampage	https://www.huffpost.com/entry/puerto-rico-water-hurricane-fiona_n_632bdfd8e4b0d12b54014e13	More than half a million people remained without water service three days after the storm lashed the U.S. territory.
8	Marina Fang	CULTURE & ARTS	2022-09-22	How A New Documentary Captures The Complexity Of Being A Child Of Immigrants	https://www.huffpost.com/entry/mija-documentary-immigration-isabel-castro-interview_n_632329aee4b000d98858dbda	In "Mija," director Isabel Castro combined music documentaries with the style of "Euphoria" and "Clueless" to tell a more nuanced immigration story.
9	Aamer Madhani, AP	WORLD NEWS	2022-09-21	Biden At UN To Call Russian War An Affront To Body's Charter	https://www.huffpost.com/entry/biden-un-russian-war-an-affront-to-bodys-charter_n_632ad9e3e4b0bfdf5e1bf5f7	White House officials say the crux of the president's visit to the U.N. this year will be a full-throated condemnation of Russia and its brutal war.

سپس برای انجام پیش پردازش های لازم ، جملات عنوان خبرهارا به کلمات جدا شده تبدیل می کنیم :

```
[6] from pyspark.ml.feature import Tokenizer, StopWordsRemover, CountVectorizer, IDF
    from pyspark.sql.functions import col

    # 1. Tokenization (جدا کردن کلمات)
    tokenizer = Tokenizer(inputCol="headline", outputCol="tokens")
    tokenized_df = tokenizer.transform(news_df)
```

index	headline	category	tokens
0	Over 4 Million Americans Roll Up Sleeves For Omicron-Targeted COVID Boosters	U.S. NEWS	over,4,million,americans,roll,up,sleeves,for,omicron-targeted,covid,boosters
1	American Airlines Flyer Charged, Banned For Life After Punching Flight Attendant On Video	U.S. NEWS	american,airlines,flyer,charged,,banned,for,life,after,punching,flight,attendant,on,video
2	23 Of The Funniest Tweets About Cats And Dogs This Week (Sept. 17-23)	COMEDY	23,of,the,funniest,tweets,about,cats,and,dogs,this,week,(sept.,17-23)
3	The Funniest Tweets From Parents This Week (Sept. 17-23)	PARENTING	the,funniest,tweets,from,parents,this,week,(sept.,17-23)
4	Woman Who Called Cops On Black Bird-Watcher Loses Lawsuit Against Ex-Employer	U.S. NEWS	woman,who,called,cops,on,black,bird-watcher,loses,lawsuit,against,ex-employer
5	Cleaner Was Dead In Belk Bathroom For 4 Days Before Body Found: Police	U.S. NEWS	cleaner,was,dead,in,belk,bathroom,for,4,days,before,body,found:,police
6	Reporter Gets Adorable Surprise From Her Boyfriend While Live On TV	U.S. NEWS	reporter,gets,adorable,surprise,from,her,boyfriend,while,live,on,tv
7	Puerto Ricans Desperate For Water After Hurricane Fiona's Rampage	WORLD NEWS	puerto,ricans,desperate,for,water,after,hurricane,fiona's,rampage
8	How A New Documentary Captures The Complexity Of Being A Child Of Immigrants	CULTURE & ARTS	how,a,new,documentary,captures,the,complexity,of,being,a,child,of,immigrants
9	Biden At UN To Call Russian War An Affront To Body's Charter	WORLD NEWS	biden,at,un,to,call,russian,war,an,affront,to,body's,charter

Stop word هارا حذف می کنیم :

```
[8] # 2. Stop Words حذف
    remover = StopWordsRemover(inputCol="tokens", outputCol="filtered_tokens")
    filtered_df = remover.transform(tokenized_df)
```

پس از اعمال پیش پردازش های لازم ، الگوریتم bag of words vectorization را با استفاده از قطعه کد زیر اجرا می کنیم :

```
# 3. Bag of Words (CountVectorizer)
cv = CountVectorizer(inputCol="filtered_tokens", outputCol="BoW_features")
cv_model = cv.fit(filtered_df)
BoW_df = cv_model.transform(filtered_df)
```

و در ادامه اهمیت کلمات را به کمک الگوریتم TF-IDF به بردار ها اعمال می کنیم :

```
from pyspark.ml.feature import Tokenizer, HashingTF, IDF

hashingTF = HashingTF(inputCol="filtered_tokens", outputCol="TF_Features", numFeatures=3000)
tfData = hashingTF.transform(BoW_df)
```

```
[11] idf = IDF(inputCol="TF_Features", outputCol="features")
     idfModel = idf.fit(tfData)
     tfidfData = idfModel.transform(tfData)
```

در مرحله ی بعد از این بردار های TF-IDF برای تشکیل درخت استفاده می کنیم . اما قبل از آن داده های مربوط به ستون category را به label تبدیل می کنیم تا بتوانیم برای آموزش درخت از آن استفاده کنیم :

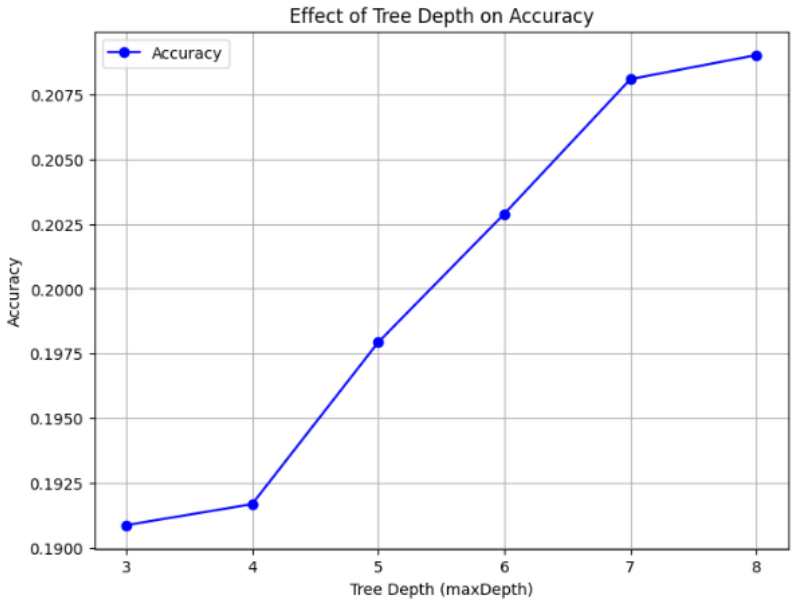
```
# ایجاد لیبل
indexer = StringIndexer(inputCol="category", outputCol="label")
final_data = indexer.fit(tfidfData).transform(tfidfData)
```

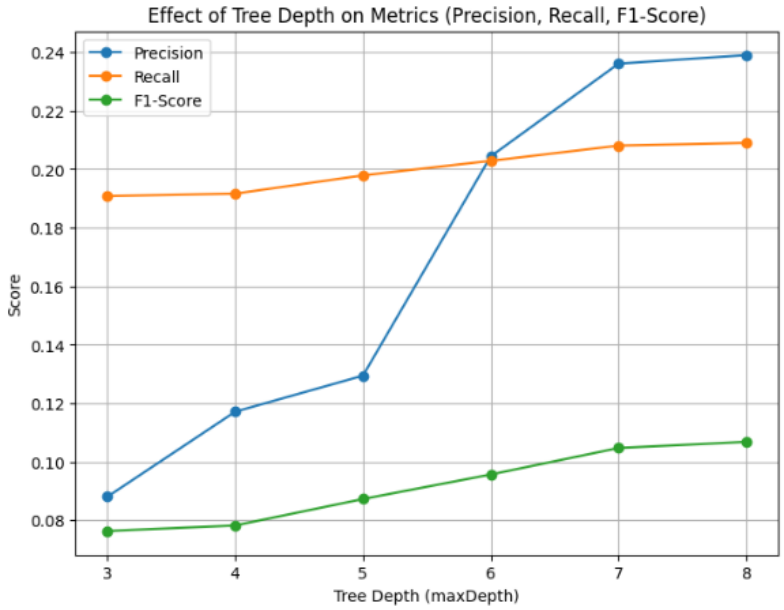
با استفاده از قطعه کد زیر درخت تصمیم خودمون رو تشکیل می دهیم :

```python
from pyspark.sql import SparkSession
from pyspark.ml.feature import StringIndexer, VectorAssembler
from pyspark.ml.classification import DecisionTreeClassifier
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
import matplotlib.pyplot as plt

# ایجاد لیبل
indexer = StringIndexer(inputCol="category", outputCol="label")
final_data = indexer.fit(tfidfData).transform(tfidfData)

# تقسیم داده‌ها به آموزش و تست
train_df, test_df = final_data.randomSplit([0.8, 0.2], seed=42)

# تعیین عمق های درخت
depths = range(3, 9)
metrics = {"accuracy": [], "precision": [], "recall": [], "f1": []}

for depth in depths:
    # تعریف مدل درخت تصمیم
    dt = DecisionTreeClassifier(labelCol="label", featuresCol="features", maxDepth=depth)

    # آموزش مدل
    model = dt.fit(train_df)

    # پیش‌بینی روی داده‌های تست
    predictions = model.transform(test_df)

    # ارزیابی دقیق
    evaluator = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction")

    # محاسبه معیارها
    accuracy = evaluator.evaluate(predictions, {evaluator.metricName: "accuracy"})
    precision = evaluator.evaluate(predictions, {evaluator.metricName: "weightedPrecision"})
    recall = evaluator.evaluate(predictions, {evaluator.metricName: "weightedRecall"})
    f1 = evaluator.evaluate(predictions, {evaluator.metricName: "f1"})

    # ذخیره نتایج
    metrics["accuracy"].append(accuracy)
    metrics["precision"].append(precision)
    metrics["recall"].append(recall)
    metrics["f1"].append(f1)
```

```python
# رسم نمودار Accuracy
plt.figure(figsize=(8, 6))
plt.plot(depths, metrics["accuracy"], marker='o', linestyle='-', color='b', label="Accuracy")
plt.title("Effect of Tree Depth on Accuracy")
plt.xlabel("Tree Depth (maxDepth)")
plt.ylabel("Accuracy")
plt.grid(True)
plt.legend()
plt.show()

# رسم نمودار Precision, Recall, F1-Score
plt.figure(figsize=(8, 6))
plt.plot(depths, metrics["precision"], marker='o', linestyle='-', label="Precision")
plt.plot(depths, metrics["recall"], marker='o', linestyle='-', label="Recall")
plt.plot(depths, metrics["f1"], marker='o', linestyle='-', label="F1-Score")
plt.title("Effect of Tree Depth on Metrics (Precision, Recall, F1-Score)")
plt.xlabel("Tree Depth (maxDepth)")
plt.ylabel("Score")
plt.legend()
plt.grid(True)
plt.show()
```

عملکرد درخت در عمق های مختلف به صورت زیر است :



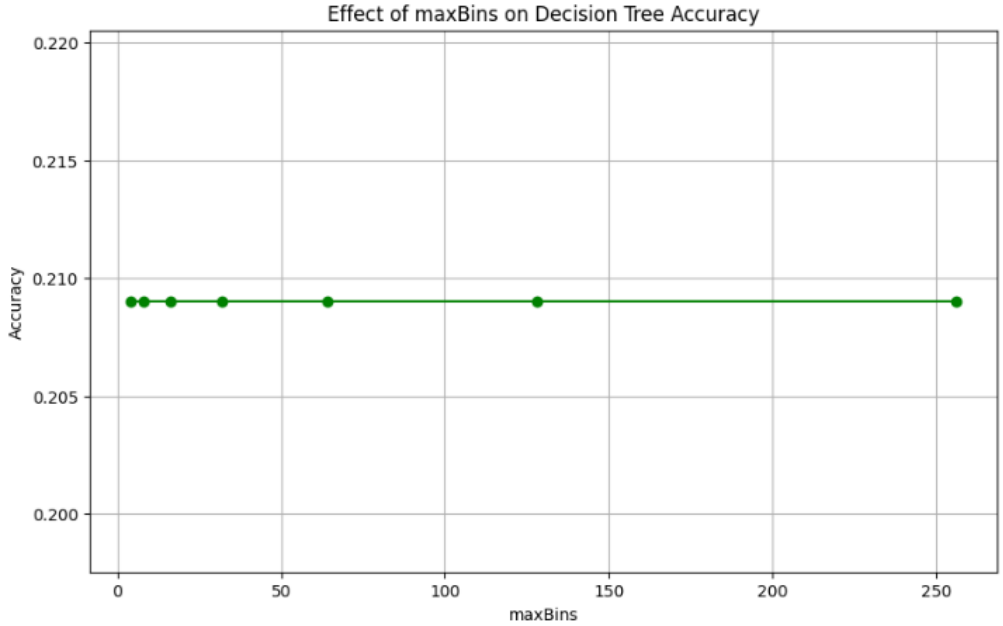


در قسمت بعدی تاثیر پارامتر maxbins روی دقت درخت را بررسی می کنیم :

```
# مقادیر maxBins برای آزمایش
bins = [4, 8, 16, 32, 64, 128, 256]

# (عمق بهینه به دست آمده از قسمت قبل
optimal_depth = 8

# متغیر برای ذخیره دقت‌ها
accuracies_bins = []

# maxBins آموزش مدل برای مقادیر مختلف
for max_bin in bins:
    dt = DecisionTreeClassifier(featuresCol="features", labelCol="label", maxDepth=optimal_depth, maxBins=max_bin)
    model = dt.fit(train_df)
    predictions = model.transform(test_df)

    evaluator = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction", metricName="accuracy")
    accuracy = evaluator.evaluate(predictions)
    accuracies_bins.append(accuracy)
    print(f"Accuracy for maxBins {max_bin}: {accuracy}")

# رسم نمودار
plt.figure(figsize=(10, 6))
plt.plot(bins, accuracies_bins, marker='o', linestyle='-', color='g')
plt.title("Effect of maxBins on Decision Tree Accuracy")
plt.xlabel("maxBins")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()
```

که متاسفانه این پارامتر روی درخت ما هیچ تاثیری ندارد .

```
Accuracy for maxBins 4: 0.2090318574771935
Accuracy for maxBins 8: 0.2090318574771935
Accuracy for maxBins 16: 0.2090318574771935
Accuracy for maxBins 32: 0.2090318574771935
Accuracy for maxBins 64: 0.2090318574771935
Accuracy for maxBins 128: 0.2090318574771935
Accuracy for maxBins 256: 0.2090318574771935
```



پارامتر maxBins معمولاً زمانی اهمیت پیدا می‌کند که داده‌های ما شامل ویژگی‌های دسته‌بندی شده (Categorical Features) یا گسسته باشند. اگر تمام ویژگی‌های ما عددی و پیوسته (Continuous) هستند، maxBins ممکن است تأثیری نداشته باشد.

همچنین ویژگی هایی که ازTF-IDF به دست می آید معمولاً پیوسته هستند. در این حالت، تغییر maxBins تأثیر زیادی نخواهد داشت.

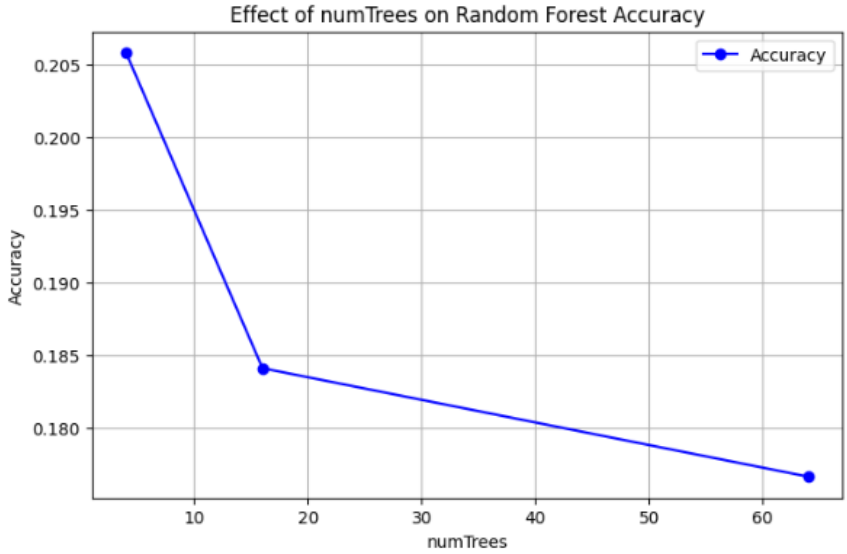
در مرحله ی بعدی یک مدل جنگل تصادفی با maxdepth بهینه ی به دست آمده در قسمت قبلی با تعداد درخت های ۴ و ۱۶ و ۳۲ طراحی می کنیم که به صورت قطعه کد زیر آن را نوشتیم :

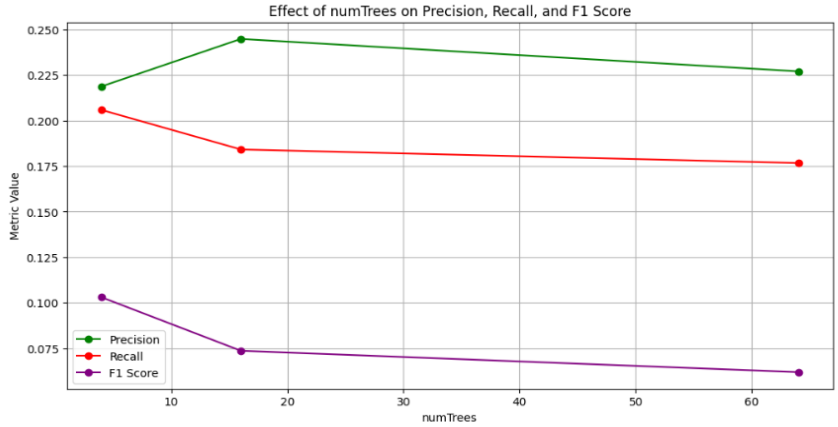
```python
from pyspark.ml.classification import RandomForestClassifier
from pyspark.ml.evaluation import MulticlassClassificationEvaluator
import matplotlib.pyplot as plt
from time import time

# numTrees مقادیر مختلف
num_trees = [4, 16, 64]

# پارامتر maxDepth بهینه شده از تمرین های قبلی
optimal_depth = 8

# Recall, Precision, F1 Score و Accuracy ذخیره مقادیر برای
metrics = {
    "accuracy": [],
    "precision": [],
    "recall": [],
    "f1": [],
    "execution_time": []
}

# numTrees محاسبه متریکها برای هر مقدار
for trees in num_trees:
    rf = RandomForestClassifier(featuresCol="features", labelCol="label", maxDepth=optimal_depth, numTrees=trees)

    start_time = time()
    model = rf.fit(train_df)
    predictions = model.transform(test_df)
    end_time = time()

    # محاسبه متریکها
    evaluator_accuracy = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction", metricName="accuracy")
    evaluator_precision = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction", metricName="weightedPrecision")
    evaluator_recall = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction", metricName="weightedRecall")
    evaluator_f1 = MulticlassClassificationEvaluator(labelCol="label", predictionCol="prediction", metricName="f1")

    metrics["accuracy"].append(evaluator_accuracy.evaluate(predictions))
    metrics["precision"].append(evaluator_precision.evaluate(predictions))
    metrics["recall"].append(evaluator_recall.evaluate(predictions))
    metrics["f1"].append(evaluator_f1.evaluate(predictions))
    metrics["execution_time"].append(end_time - start_time)

    print(f"numTrees={trees} -> Accuracy: {metrics['accuracy'][-1]}, Precision: {metrics['precision'][-1]}, Recall: {metrics['recall'][-1]}, F1: {metrics['f1'][-1]}, Time: {metrics['execution_time'][-1]}")
```

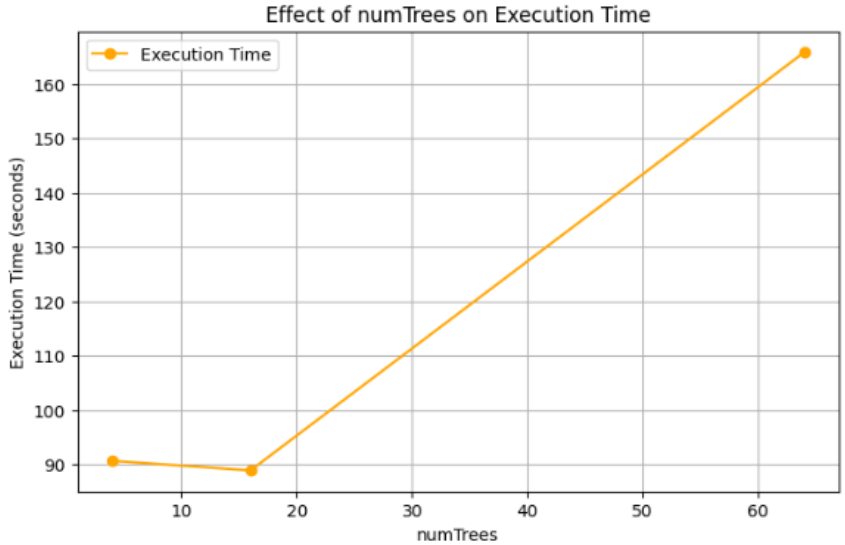
```python
# Accuracy رسم نمودار
plt.figure(figsize=(8, 5))
plt.plot(num_trees, metrics["accuracy"], marker='o', linestyle='-', color='b', label="Accuracy")
plt.title("Effect of numTrees on Random Forest Accuracy")
plt.xlabel("numTrees")
plt.ylabel("Accuracy")
plt.grid(True)
plt.legend()
plt.show()

# Precision, Recall و F1 Score رسم نمودارهای
plt.figure(figsize=(12, 6))
plt.plot(num_trees, metrics["precision"], marker='o', linestyle='-', color='g', label="Precision")
plt.plot(num_trees, metrics["recall"], marker='o', linestyle='-', color='r', label="Recall")
plt.plot(num_trees, metrics["f1"], marker='o', linestyle='-', color='purple', label="F1 Score")
plt.title("Effect of numTrees on Precision, Recall, and F1 Score")
plt.xlabel("numTrees")
plt.ylabel("Metric Value")
plt.grid(True)
plt.legend()
plt.show()

# رسم نمودار زمان اجرا
plt.figure(figsize=(8, 5))
plt.plot(num_trees, metrics["execution_time"], marker='o', linestyle='-', color='orange', label="Execution Time")
plt.title("Effect of numTrees on Execution Time")
plt.xlabel("numTrees")
plt.ylabel("Execution Time (seconds)")
plt.grid(True)
plt.legend()
plt.show()
```

نتایج به دست آمده از اجرای این الگوریتم به شرح زیر است :

```
numTrees=4 -> Accuracy: 0.20578401872283517, Precision: 0.21870417046399412, Recall: 0.20578401872283517, F1: 0.10304829022487866, Time: 90.61253833770752
numTrees=16 -> Accuracy: 0.1841237999713426, Precision: 0.24474500850138048, Recall: 0.1841237999713426, F1: 0.07371230023878995, Time: 88.83556795120239
numTrees=64 -> Accuracy: 0.17667287577016766, Precision: 0.22696906443520423, Recall: 0.17667287577016763, F1: 0.06199190560830413, Time: 165.87679290771484
```







در بخش آخر میخواهیم تعداد هسته های پردازشی اسپارک را افزایش بدهیم تا تاثیر آن را روی سرعت خوشه بندی مشاهده کنیم . به منظور این کار ، جلسه ی اسپارک را با قطعه کد زیر شروع می کنیم :

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("OptimizedKMeans") \
    .config("spark.executor.memory", "8g") \  # حافظه بیشتر برای Executor ها
    .config("spark.executor.cores", "4") \    # افزایش تعداد هسته‌ها
    .config("spark.sql.shuffle.partitions", "200") \  # کاهش تعداد پارتیشن‌ها
    .config("spark.default.parallelism", "200") \  # تعداد وظایف همزمان
    .getOrCreate()
```

که تعداد هسته های پردازشی اسپارک را به عدد ۴ افزایش می دهد .

این کار تغییر محسوسی روی سرعت و نتایج به دست آمده نزاشت و فقط در حد چند دقیقه رسیدن به نتایج را سریعتر کرد .